

James Fox
Direct: 815-666-3405

LinkedIn: <https://www.linkedin.com/in/james-p-fox747/>
Email: jpfox19998@gmail.com (Preferred contact method)

Summary: Mechanical Engineer with hands-on experience in medical device and pharmaceutical manufacturing environments, supporting quality engineering, validation, automation, and process improvement initiatives. Experienced in CSV, I/O/PQ, FAT/SAT execution, CAPA, non-conformance investigations, DFMEA/PFMEA, and root cause analysis within regulated environments. Proven track record of leading cross-functional projects, improving production reliability, and reducing manufacturing defects. Seeking an entry to mid-level engineering or technician role in quality, manufacturing, mechanical, or project engineering.

Education: Illinois Institute of Technology, Chicago, IL
Degree: Bachelor of Science, Mechanical Engineering 2017-21

Top Skills: Quality Assurance and lean/6S methodologies, Technical Drawing, Drafting and Writing, Internal-Company and Cross-Company Project Management, Mechanical Design Fabrication and Analysis, and Research and Development.

CAD Programs: Creo, Autodesk Inventor, AutoCAD, and Solid Works

Coding Languages: Python, Mathematica, MATLAB, Simulink.

PLM: SAP

Employment:

Vectro Solutions, Travel Position

December 2025-February 2026

Application QA Validation Engineer (Short-Term Project, Contract)

- Travel position that required me to be out of state the duration of the project. This project was for a client in the pharma industry.
- Developed Computer System Validation (CSV) protocols through the use of the vendor provided materials, client user requirements and modifications, and through running performance tests on the system itself.
 - After its initial drafting would then perform a “dry run” of the protocol to improve its accuracy.
- Executed CSV and IQQ protocols on client equipment and systems to validate it for use in commercial runs.
 - Developed discrepancy reports for issues that occurred during validation activities.
 - Worked with client staff to troubleshoot and resolve these discrepancies.
 - Updated the CSV and IQQ protocols to reflect these discrepancies for future protocol executions.

Jabil

July 2025-October 2025

Validation Engineer (Short-Term Project, Contract Agency)

- Specialized in ionizer installation and validation for injection molded plastic components for the medical industry.
- Performed production line upgrades with the installation of new fan and emitter force ionizers. This included modifying existing electrical components on the line to adapt to the new components.
- Worked hands-on with maintaining and breaking down the injection molding machines used on the manufacturing floor.
- Monitored Ionizer activity daily, utilizing control charts for QC and maintenance purposes.
- Assisted in the selection and buying of new equipment for customer projects.
- Worked with the QA department for CAPA projects to lead upgrades to production lines and product quality inspections.

Abbott, Greater Chicago Land Area

February 2025-May 2025

Field Service Engineer (Short-Term Project, Contract Agency)

- Completed FSE training for the Alinity M system, its Amp Detect units, and other Abbott electro-mechanical medical devices.
- Completed refurbishments on Abbott devices, primarily the ADUs, under various regulatory standards including CE and UL.
- The refurbishment process followed the principles of PDCA and included:
 - Diagnosing the issue that caused the unit to be at fault and decontaminating the unit.
 - Performing the necessary repairs to resolve the issue. This could include fixing software and replacing/repairing subcomponents of the unit including the power conversion boards, thermal conversion boards, and main boards.
 - Performing test runs using the refurbished units in the Alinity M.

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- o Completing all necessary paperwork, updates to SAP, and collecting all required data per company standard

Medline, Arlington Heights, IL

July 2024-February 2025

Associate Quality Engineer, Respiratory (Contract Agency)

- Completed investigations into customer complaints and supported other QEs in their investigations.
- Completed non-conformance investigations and reports for medical manufacturing.
- These investigations were completed in SAP through DFMEA, PFMEA, GD&T, RCA, and customer interviews.
- Reviewed, corrected, and submitted DHRs for product release. Instructed other QEs on the process as well.
- Conducted Incoming inspections for raw materials and investigations for incoming completed goods.

AbbVie, North Chicago, IL

April 2023-March 2024

Project Engineer, Automation (Contract Agency)

- Led supplier qa investigations into the automation equipment on the production lines. As a result, potential downtime due to lead time for spare components and/or incorrect maintenance due to faulty documentation was prevented.
 - o Discovered trends of errors in official documentation, signs of missing required documentation, errors in machine drawings, missing machine drawings, and missing spare components while leading a Site Acceptance Test which led to the investigations.
 - o Led the execution of corrective actions to any issues found during the investigations.
 - o The SAT was completed on time. The rest of the automation equipment and documentation was corrected before any errors could cause downtime.
- Executed and led SATs, I/O/PQs, validation runs, and other tests per company standards and procedures.
 - o Scheduled between internal teams and led my own to complete the runs with a sense of urgency.
- Coordinated work/projects on the manufacturing floor between the production department, the automation department, and service providers.
- Led database management and corrections for automation formats.
 - o Created a warning system for high-risk formats to limit downtime from this database.
- Led training sessions for the use and maintenance of new equipment.
- Led and completed upgrades and/or repairs to automation equipment and programs used in manufacturing.
- Created a new database using Microsoft SharePoint for the projects being worked on by the automation team, led weekly meetings to discuss the projects, and instructed my team on how to use the database.
- Represented the automation department in weekly international cross company project planning.

10x Microstructures, Wheeling, IL

June 2022-January 2023

Mechanical Engineer

- Led project utilizing the PDCA cycle and the Ishikawa Diagram to reduce the rate of nanoscopic errors on electroformed contact molds by 70%. Through DFMEA and PFMEA each step of the production process was tested and verified for where errors were occurring. During the process, electromechanical devices created for measurement.
 - o Investigated the first step of the process, the fabrication of silicon molds used in electroforming.
 - Created a device to map the silicone for any errors so any defects after its production could be caught and avoided before it could be used in the next steps.
 - o Investigated the electroforming and bonding steps next utilizing GD&T and experiment design to correct the issues in these steps.
 - Analyzed completed molds and the tools used to make them using white light profilometry and other measurement tools and techniques to determine the root cause of the errors.
 - Found that they failed due to how the molds were formed, bonded off-center, and cleaned improperly based on the data collected.
 - o Redesigned and created new work instructions for the electroforming process to eliminate particulates and wrinkles in the silicon that occur during the preparation step after designing after successfully experimenting with the clean room design and silicon lay down process.

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- Created a device to measure the off-center value of the completed molds and determined that the issue came from the apparatuses used for the bonding process.
 - From the information gathered, analyzed and redesigned the bonding apparatuses to eliminate the off-center value of the completed molds.
- Redesigned the post-machining cleaning process to eliminate foreign particles on the surface of the completed molds through experimentation with different cleaning techniques and chemicals.
- Created a new machine to measure light passing through resin tubing for continuous glucose monitoring to make the measurement process more efficient and quicker.
 - For this project also developed programs for the on sight CNC machine to cut through the film without damaging the remaining material as well as assisting in the purchasing of new CNC equipment.
- Designed the nanoscopic pattern of the blood oxygenation tissue of an artificial lung and led the analysis of the tissue utilizing SEM and white light profilometry.
- Coached other staff on the use of the newly developed machines and techniques to perform quality assurance analysis on manufactured components, to assist in the bonding process, and update the companies PLM database for the molds.

Graphic Packaging International, Chicago, IL
Reliability Engineer, Short-Term Project, Contract Agency

August 2021-November 2021

- Performed process capability analysis and created a process flow diagram to determine downtime for services and repairs to production machinery and printer equipment.
- From this, with production, I created the most optimal repair schedule for the machines and reduced the average plant downtime by 15%
- Created and presented Gantt charts for larger maintenance projects, held daily meetings and Gemba walks to discuss them with the other plant departments as well as working with them on their projects.

Personal Projects:

Antimatter Interferometer, Chicago, IL, Role: Project Manager.

January 2021-May 2021

- Managed a group of 40 people divided into 8 teams with the purpose of furthering the construction of an interferometer to determine the gravitational properties of antimatter.
- Led weekly meetings to track progress and to schedule assistance, work, and projects for the teams.
- Assisted teams individually to gain an understanding of their specialization and be able to be a member when needed.

Thunderstep, Chicago IL, Role: Lead Designer

January 2021-May 2021

- The Thunderstep was a shoe that used magnetic induction to store the power from the motion of stepping for later use.
- Developed a system of pistons to keep the magnets and copper coils in place during regular use of the shoe.
- Created the technical documentation for the Thunderstep and Led the fabrication and testing of the prototype.
 - Fabrication involved creating components using CNC machining and 3D printing, as well as utilizing a soldering iron to assemble the circuitry responsible for charging the battery.